

Research article for the Surreal project

Omnific Integers and Omnific–Diophantine Geometry

Retractions, rigidity, definability,
and infinite families

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Abstract

The omnific integers form a discretely ordered subring of the surreal field, but their arithmetic is neither ordinary integer arithmetic nor a routine nonstandard enlargement of it. We organize their theory around the split ring homomorphism that takes a normal form to its constant coefficient. This gives a precise transfer theorem for polynomial equations with ordinary integer coefficients, a classification of finite quotients, and a complete solution criterion for linear systems.

The main Diophantine results distinguish existence from the structure of solutions. Nonzero fibers of nondegenerate binary forms and number-field norm forms have no new omnific points. In particular, Pell equations admit only ordinary-integer solutions. In contrast, every represented nonzero level of an indefinite nondegenerate quadratic form in at least three variables has explicit infinite polynomial families through each integer point. We prove the resulting quadratic-level dichotomy.

Combining Pell rigidity with the classical three-square theorem produces a single quartic polynomial, with five auxiliary variables independent of the tuple length, defining exactly the ordinary integers inside the omnific integers. Further results concern non-atomic factorization, a common-divisor phenomenon peculiar to the full surreal class, projective denominator clearing, primitive Pythagorean triples, homogeneous cones, and obstructions at infinity. All central deductions receive proofs. Historical background, imported theorems, proposed contributions, and unverified formalization work are explicitly separated.

Prepared with OpenAI ChatGPT. This is a mathematical research draft, not a claim of peer review or proof-assistant verification. The included program checks finite identities and examples, not the entire theory. Repository inspection is pinned to the commit recorded in [Appendix B](#).

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1 Scope, results, and mathematical conventions

Omnific integers are the surreal numbers with no negative exponents in Conway normal form and with an ordinary integer constant coefficient. The definition is classical, as are their discreteness and integer-part property [1, 2, 3]. This article develops their arithmetic with a particular emphasis on polynomial equations.

Three questions must be kept separate. Does an equation have any solution? Does it have a genuinely infinite solution? Does it have such a solution with a primitivity condition? For example, every system of equations over $\mathbb{Z}$ has an omnific solution exactly when it has an ordinary integer solution. Nevertheless, the equation $X^n + Y^n = Z^n$ has positive infinite omnific solutions for every ordinary $n \geq 2$. There is no contradiction: the constant coefficients of these solutions are all zero.

We write $A = \mathbf{Oz}$. The ideal $\mathcal{J}$ consists of the normal forms supported on strictly positive exponents; coefficients in $\mathcal{J}$ may have either sign. The central decomposition is

$$A = \mathbb{Z} \oplus \mathcal{J}, \quad \text{ct} : A \longrightarrow \mathbb{Z}, \quad \text{ct}(a + b) = \text{ct}(a) + \text{ct}(b), \quad \text{ct}(ab) = \text{ct}(a) \text{ct}(b). \quad (1.1)$$

It is a splitting of additive groups and a ring retraction, not a direct product decomposition of rings. In particular, $\mathcal{J}$ has no identity.

1.1 Principal results

The structural results in Sections 2–4 explain why ordinary congruences see only the constant coefficient, although the fraction field of the full class A is all of $\mathbf{No}$. The main Diophantine statements are the following.

- (1) **Equational transfer.** For every finite system over $\mathbb{Z}$, $V(A)$ is nonempty exactly when $V(\mathbb{Z})$ is nonempty (theorem 5.1). This does not include inequations or order conditions.
- (2) **Constant-product rigidity.** A nonzero constant product of normal forms with no negative exponents has only constant factors. This forces all omnific points on nonzero binary-form and number-field norm fibers to be ordinary integer points (theorems 6.3 and 6.7).
- (3) **Quadratic-level dichotomy.** For $q \in \mathbb{Z}[X_1, \dots, X_n]$ quadratic homogeneous and $0 \neq c \in \mathbb{Z}$, all solutions are ordinary in the nondegenerate definite case and in nondegenerate dimension at most two. In every other case, each integer point lifts to an injectively parameterized infinite family (theorem 7.1).
- (4) **A quartic definition of $\mathbb{Z}^n$.** The equation

$$(U^2 - 2V^2 - 1)^2 + \left(U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2 = 0$$

with five existentially quantified auxiliary variables defines exactly $\mathbb{Z}^n$ inside A^n (theorem 8.2).

- (5) **Primitive and projective phenomena.** There are infinite unimodular Pythagorean triples. On the other hand, a constant real projective direction has a primitive omnific representative precisely when it is rational (proposition 9.4 and theorem 9.5).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. The constant-term formalism is classical; several consequences are

natural applications of it. The quadratic classification and explicit quartic guard are the most distinctive constructions developed in this article. The inspected sources did not provide these statements in this form. No exhaustive historical-priority claim is made.

1.2 Class size and finite algebra

The full surreal field and A are proper classes. We work in a customary class-theoretic setting, such as NBG with the usual background for surreal normal forms. Each individual surreal has a *set-sized* support. Every polynomial, system, matrix, or tuple in this article has ordinary finite size. Quantifiers over A are understood as class quantifiers expressed through their defining set-theoretic formulas, not as an appeal to a set containing all surreals.

Quotients by class ideals are read through their quotient maps and congruence relations. In particular, $A/\mathcal{J} = \mathbb{Z}$ means that ct is onto $\mathbb{Z}$ and $x - y \in \mathcal{J}$ exactly when $\text{ct}(x) = \text{ct}(y)$; a residue is represented by its ordinary integer value. Similarly, A/nA is represented by the ordinary residue ring $\mathbb{Z}/n\mathbb{Z}$. No proper-class coset is used as an element of a set. Ideal generation uses finite sums, even when the family of generators is a class. The notation $\text{Frac}(A)$ below is interpreted through fractions inside **No**, not through a set of all pairs from A .

Most proofs can instead be carried out in a fixed set-sized Hahn field. We explicitly identify the arguments that need the stronger fact that *every set* of surreal exponents has upper bounds and every set of positive surreal exponents has a strictly positive lower bound. One must not silently assume these properties for a fixed ordered exponent group.

The notation ω^γ always means the Conway monomial. Its multiplication is $\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$, where $+$ is surreal addition, not ordinary ordinal addition. Exponent supports are reverse well ordered. In the frequent Hahn convention $t = \omega^{-1}$, all exponent signs are reversed.

1.3 Relationship with the repository and literature

Integration status. This is manuscript 01 of the three-source Diophantine report placed in repository commit `be06fc8`. The present integration covers its basic normal-form, residue and global-support arguments, compared with manuscripts 02 and 05. Additional consequences from those sources are identified below. Their later Diophantine and differential results have not yet all been integrated. The maintained `README.md` and `RECONCILIATION.md` record this boundary; statements later in this base text retain their original scope and attribution. Repository placement does not constitute proof review or Lean formalization.

The repository's documentation distinguishes normal-form algebra, fixed Hahn workspaces, and the full surreal carrier. Its existing trigonometry material already uses the omnific integer part and residue arithmetic [10]. We retain those distinctions and develop a separate Diophantine treatment rather than identifying A with a finite-surreal valuation ring.

The main external sources are Conway and Gonshor for the construction and normal forms [1, 2], L'Innocente–Mantova for modern factorization theory [3], Ehrlich–Kaplan for the initial integer-part context [4], and Jeřábek for the general residue theory of $\mathbb{Z}$ -groups [5]. The three-square theorem and MRDP theorem are imported with explicit attribution at their points of use.

2 The normal-form ring and its integer-part structure

2.1 Normal forms and three related rings

We use the standard normal-form theorem: each surreal has a unique expression

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad r_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where $S \subseteq \mathbf{No}$ is a reverse-well-ordered set. Conversely such a normal form represents a surreal. Addition is coefficientwise, multiplication is the well-defined Hahn convolution, and the sign of a nonzero normal form is the sign of its leading coefficient [1, 2].

Definition 2.1. Let

$$\begin{aligned} \mathcal{J} &= \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\}, \\ \mathcal{B}_{\mathbb{R}} &= \mathbb{R} \oplus \mathcal{J}, \\ A = \mathbf{Oz} &= \mathbb{Z} \oplus \mathcal{J}, \\ \mathcal{B}_{\mathbb{C}} &= \mathcal{B}_{\mathbb{R}}[i] \subseteq \mathbf{No}(i). \end{aligned}$$

The zero series belongs to $\mathcal{J}$. For an element of any of these rings, $\text{ct}(x)$ denotes its coefficient at exponent zero. For $x = a + ib \in \mathcal{B}_{\mathbb{C}}$, this is $\text{ct}(a) + i \text{ct}(b)$.

These are growth-exponent conventions. With $t^\gamma = \omega^{-\gamma}$, the supports of $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{B}_{\mathbb{C}}$ have only nonpositive t -exponents, and $\mathcal{J}$ has only strictly negative t -exponents. This is a restriction on every supported exponent, not just the leading one. In the complex ring, the kernel of ct is $\mathcal{J} + i\mathcal{J}$, an ideal of $\mathcal{B}_{\mathbb{C}}$; it is not the real ideal $\mathcal{J}$ alone.

Thus $\sqrt{2}\omega$, $\omega/17$, $-\omega^\omega + 3$, and $\sum_{k \geq 1} \omega^{1/k}$ are omnific integers. The numbers $\omega + \frac{1}{2}$ and $\omega + \omega^{-1}$ are not. The word “integer” restricts the constant coefficient and excludes infinitesimal tails; it does not require all coefficients to lie in $\mathbb{Z}$.

For every positive surreal exponent γ , even an infinitesimal one, the Conway monomial ω^γ exceeds every ordinary positive real number. This is a normal-form order fact about the omega-map; replacing that map by an unstated analytic notion of exponentiation would invalidate the support arguments below.

Proposition 2.2. $\mathcal{B}_{\mathbb{R}}$, $\mathcal{B}_{\mathbb{C}}$, and A are integral domains. The class $\mathcal{J}$ is an ideal of $\mathcal{B}_{\mathbb{R}}$ and A , and is an $\mathbb{R}$ -vector space. Constant-coefficient extraction is a surjective ring homomorphism from $\mathcal{B}_{\mathbb{R}}$ to $\mathbb{R}$, from $\mathcal{B}_{\mathbb{C}}$ to $\mathbb{C}$, and from A to $\mathbb{Z}$. In the last case its kernel is $\mathcal{J}$, and the inclusion $\mathbb{Z} \hookrightarrow A$ is a section.

Proof. Nonnegative exponents remain nonnegative under addition of exponents. A strictly positive exponent plus a nonnegative exponent is strictly positive. These observations and Hahn convolution prove the ring and ideal claims. In a product of two supported normal forms, exponent zero can arise only as $0 + 0$. Therefore $\text{ct}(xy) = \text{ct}(x) \text{ct}(y)$. The additive claim is immediate. Surjectivity and the section are supplied by constant series. All rings embed in the relevant field, so have no zero divisors. $\square$

Remark 2.3 (A frequently misleading analogy). The map ct is not a ring homomorphism on all of $\mathbf{No}$: $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$ but $\text{ct}(1) = 1$. Also $\mathcal{B}_{\mathbb{R}}$ is not the ring of finite surreals. The latter permits negative exponents but no positive exponents; $\mathcal{B}_{\mathbb{R}}$ has the opposite support restriction. In particular, taking standard part of a finite element and taking constant coefficient of an omnific integer are different constructions, although both are multiplicative on their own respective domains. The retraction is not order preserving: $\omega - 1 > 0$ but $\text{ct}(\omega - 1) = -1$. Thus its use in equational arguments does not transport inequalities.

2.2 Leading degree, units, and finite elements

For nonzero $x \in \mathcal{B}_{\mathbb{C}}$, combine real and imaginary coefficients and put

$$\deg_{\omega}(x) = \max \operatorname{supp}(x), \quad \operatorname{lc}(x) = [\omega^{\deg_{\omega}(x)}]x.$$

The degree lies in $\mathbf{No}^{\geq 0}$. It is not an ordinal and not a birthday. For the same nonzero form in t -notation, $v(x) = -\deg_{\omega}(x)$.

Lemma 2.4. *For nonzero $x, y \in \mathcal{B}_{\mathbb{C}}$,*

$$\deg_{\omega}(xy) = \deg_{\omega}(x) + \deg_{\omega}(y), \quad \operatorname{lc}(xy) = \operatorname{lc}(x) \operatorname{lc}(y).$$

An element of $\mathcal{B}_{\mathbb{C}}$ of degree zero is an ordinary complex constant.

Proof. If γ and δ are the respective largest exponents, every contributing exponent is at most $\gamma + \delta$. Equality requires both exponents to be largest: making either smaller makes their sum smaller. Thus the coefficient at $\gamma + \delta$ is the nonzero product of leading coefficients. Finally, support in $[0, 0]$ contains only zero. $\square$

Proposition 2.5. *The unit groups are*

$$\mathcal{B}_{\mathbb{R}}^{\times} = \mathbb{R}^{\times}, \quad \mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}, \quad A^{\times} = \{1, -1\}.$$

Moreover, the omnific integers bounded in absolute value by some ordinary real number are exactly the ordinary integers. Consequently A is discretely ordered, with least positive element 1.

Proof. If $xy = 1$ in $\mathcal{B}_{\mathbb{C}}$, nonnegativity and additivity of degree force both degrees to be zero. This gives the first two unit statements; in A the constant units are precisely the units of $\mathbb{Z}$.

A nonconstant element of A has positive leading exponent, so its absolute value exceeds every positive real number by normal-form order. A constant element of A is an integer. Hence no element lies strictly between 0 and 1, and every positive element is at least 1. $\square$

Notice that $\omega/2$ is an omnific integer even though $1/2$ is not. This is no conflict with discreteness: $\omega/2$ is infinite, and there are still no omnific integers in the interval $(0, 1)$.

2.3 The exact omnific floor

Every surreal admits a unique decomposition

$$x = P + r + \varepsilon, \quad P \in \mathcal{J}, \quad r \in \mathbb{R}, \quad \operatorname{supp}(\varepsilon) \subseteq \mathbf{No}^{<0}.$$

Here ε is infinitesimal, possibly zero.

Theorem 2.6 (Integer-part formula). *There is exactly one $a \in A$ such that $a \leq x < a + 1$. It is*

$$\lfloor x \rfloor_A = \begin{cases} P + \lfloor r \rfloor, & r \notin \mathbb{Z} \text{ or } \varepsilon \geq 0, \\ P + r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0. \end{cases} \quad (2.2)$$

Proof. For $r \notin \mathbb{Z}$, its distances from the two adjacent integers are strictly positive real numbers. The infinitesimal ε cannot cross either boundary. If r is an integer, the sign of ε determines on which side of r the number lies, and its magnitude is less than 1. This proves the displayed inequalities. Two possible floors would differ by an omnific integer strictly between -1 and 1 , so would coincide by discreteness. $\square$

For example,

$$\lfloor \omega - \omega^{-1} \rfloor_A = \omega - 1, \quad \lfloor \omega + \frac{1}{3} - \omega^{-1} \rfloor_A = \omega.$$

Merely discarding negative exponents gives the wrong answer at a negative infinitesimal displacement from an integer.

Provenance. The integer-part assertion is classical. Ehrlich–Kaplan prove the stronger initiality characterization: for an initial subfield K of $\mathbf{No}$, $\mathbf{Oz} \cap K$ is its unique initial integer part [4, Lemma 11.1]. No alternative integer part is being introduced here.

2.4 A common set-sized workspace

Lemma 2.7 (Localizing the input supports). *Every set-sized family of real or complex surreal normal forms belongs to a common Hahn field with a set-sized divisible ordered exponent group $H \subseteq (\mathbf{No}, +)$. One may take the rational span of the union of their growth-exponent supports. The nonnegative-support subrings in this field are the intersections with $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{B}_{\mathbb{C}}$, respectively.*

Proof. The support union is a set; all its finite rational linear combinations form a set-sized divisible ordered subgroup H . A normal form with support in H is a Hahn series over H with the growth order reversed. Hahn field operations, including inverses, preserve this exponent group. Intersecting with the nonnegative growth-support condition gives the stated subrings. Real and imaginary parts use the same group. $\square$

This is manuscript 05’s workspace lemma, an application of the collection’s normal-form localization principle. It does not promise that a later support bound lies in the chosen H : the common-divisor and clearing arguments in Section 4 may require enlarging it.

3 Divisibility, residues, and finite quotients

3.1 Division by ordinary integers

Theorem 3.1. *For every nonzero ordinary integer n ,*

$$nA = \mathcal{J} + n\mathbb{Z}, \quad n \mid a \iff n \mid \text{ct}(a), \quad A/nA \cong \mathbb{Z}/n\mathbb{Z}. \quad (3.1)$$

In particular,

$$\bigcap_{n \geq 1} nA = \mathcal{J}, \quad \bigcap_{k \geq 1} p^k A = \mathcal{J} \quad \text{for every ordinary prime } p. \quad (3.2)$$

Proof. Multiplication by a nonzero real scalar preserves $\mathcal{J}$ and is invertible on $\mathcal{J}$. Thus $n\mathcal{J} = \mathcal{J}$. If $a = m + j$, then $a/n = m/n + j/n$ lies in A exactly when m/n is an integer. Reduction of m modulo n gives the quotient isomorphism. Finally, the only ordinary integer divisible by all positive integers, or by every power of one prime, is zero. $\square$

This theorem solves every ordinary congruence. For instance, $\sqrt{2}\omega + 14$ is even and is congruent to 2 modulo 3. The whole infinite part is invisible to these tests.

The statement does *not* extend to arbitrary omnific divisors. For example, $\text{ct}(\omega + 2)$ divides $\text{ct}(2)$ in $\mathbb{Z}$, but $\omega + 2$ does not divide 2 in A . A product equal to the nonzero constant 2 can have only constant factors by lemma 2.4.

Corollary 3.2 (Mixed gcds with an ordinary integer). *Let $0 \neq n \in \mathbb{Z}$, $x \in A$, and let $d = \gcd(n, \text{ct}(x)) > 0$ be the ordinary positive gcd. Then the ideal generated by n and x is dA . Consequently d is a gcd of n and x in A . Every ordinary positive prime p is a prime element of A , and pA is maximal.*

Proof. Write $x = m + j$, where $m = \text{ct}(x)$ and $j \in \mathcal{J}$. Since $j/n \in \mathcal{J} \subseteq A$, the identity $x - n(j/n) = m$ gives $(n, x) = (n, m)$. Ordinary Bézout coefficients and the divisibilities $d \mid n, m$ give $(n, m) = dA$. Every common divisor divides the Bézout combination d , while d divides both generators, so it is a gcd. Finally $A/pA \cong \mathbb{Z}/p\mathbb{Z}$ is a field; hence pA is maximal and prime. $\square$

This adds manuscript 02's mixed-gcd statement and the ordinary-prime consequence also present in manuscript 05. It concerns a pair with one nonzero ordinary integer; arbitrary pairs need not have a gcd.

Corollary 3.3. *Every proper ideal $I \subset A$ whose quotient has positive characteristic is nA for a unique ordinary $n \geq 2$. Such an ideal is prime exactly when n is an ordinary prime. Every nonzero finite quotient of A is therefore a ring $\mathbb{Z}/n\mathbb{Z}$.*

Proof. If $m \cdot 1 \in I$ with $m \neq 0$, then $\mathcal{J} = m\mathcal{J} \subseteq I$. Thus I is the inverse image of an ideal of $A/\mathcal{J} = \mathbb{Z}$. Its image is $n\mathbb{Z}$ for some positive n , and properness gives $n \geq 2$. Equation (3.1) identifies its inverse image with nA . Primality is equivalent to primality of $n\mathbb{Z}$. A finite nonzero unital ring has positive characteristic, so the last assertion follows. $\square$

More generally, every unital homomorphism from A to a finite ring kills $\mathcal{J}$: if the target has characteristic m , write $j = m(j/m)$. Its image is therefore the cyclic subring generated by 1. This describes maps into finite rings as well as finite quotients.

3.2 Completions and canonicity

All finite-index ideals have just been classified, so the profinite completion, interpreted as the set-sized inverse limit of these quotients, is

$$\widehat{A} \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}. \quad (3.3)$$

The completion map is the composite $A \xrightarrow{\text{ct}} \mathbb{Z} \rightarrow \widehat{\mathbb{Z}}$, with kernel $\mathcal{J}$. Similarly,

$$\varprojlim_k A/p^k A \cong \mathbb{Z}_p$$

has kernel $\mathcal{J}$. Thus these congruence topologies on A are not Hausdorff. The corresponding separated quotient is $A/\mathcal{J} = \mathbb{Z}$; completion does not preserve the infinite part. The first inverse system is indexed by positive ordinary integers ordered by divisibility, with reduction maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ when $m \mid n$. The second uses the chain of powers of the fixed prime p . These are set-sized diagrams of ordinary residue rings. The compatible quotient identifications above justify the displayed limits without forming a set of class ideals or a set of proper-class cosets.

Proposition 3.4. *The constant-term map is the unique unital ring homomorphism $A \rightarrow \mathbb{Z}$. Every unital endomorphism $f : A \rightarrow A$ preserves $\mathcal{J}$ and satisfies $\text{ct} \circ f = \text{ct}$.*

Proof. For a homomorphism $h : A \rightarrow \mathbb{Z}$, the integer $h(j)$ is divisible by every positive integer whenever $j \in \mathcal{J}$. Hence $h(j) = 0$. Since h fixes ordinary integers, $h = \text{ct}$. An endomorphism preserves divisibility by each ordinary integer, so preserves their common divisibility kernel (3.2). It fixes $\mathbb{Z}$ pointwise, which proves the final identity. $\square$

Provenance. The divisible-kernel and residue-map perspective belongs to classical $\mathbb{Z}$ -group theory; see [5, Sections 2 and 4]. Here it has an especially concrete ring-theoretic realization because ct is multiplicative. The finite-quotient and completion statements above are proved directly rather than imported as additional assumptions.

4 Global algebra: common divisors and the full fraction field

The next results use all surreal exponents, not merely the exponents in one fixed Archimedean or finite-rank group. This is an important source of arithmetic differences between full omnific integers and restricted series rings.

4.1 One monomial divides a whole set

Lemma 4.1 (Set-sized exponent bounds). *If $E \subset \mathbf{No}^{>0}$ is a set, there is a surreal $\epsilon > 0$ such that $\epsilon < e$ for every $e \in E$. Every set of surreals also has a strict upper bound.*

Proof. A surreal separating the small cut $\{0 \mid E\}$ gives the first assertion. A separator of $\{E \mid \emptyset\}$ gives the second. The assertions also hold for the empty set. These are applications of the standard set-cut property, not an assertion of an infimum or supremum. $\square$

Theorem 4.2 (Common monomial divisibility). *For every set $S \subseteq \mathcal{J}$ there is $\epsilon > 0$ such that*

$$\omega^\epsilon \mid s \quad \text{and} \quad s/\omega^\epsilon \in \mathcal{J} \quad (s \in S).$$

In particular, each nonzero element of $\mathcal{J}$ is a product of two nonunits of A .

Proof. The union of the supports of the elements of S is a set E of strictly positive surreal exponents. Choose ϵ as in lemma 4.1. Dividing by ω^ϵ shifts every exponent e to $e - \epsilon > 0$, preserving the support order and its set size. The quotient is therefore in $\mathcal{J}$. For a nonzero s , both factors are nonzero elements with positive degree, and neither is a unit. $\square$

Even the series $\sum_{k \geq 1} \omega^{1/k}$ has such a divisor. The positive number ϵ may be infinitesimal relative to every real $1/k$. This explains why the proof cannot just be copied into a Hahn ring with exponent group $\mathbb{R}$.

Corollary 4.3. *The ideal $\mathcal{J}$ is idempotent, $\mathcal{J}^2 = \mathcal{J}$, and it is not generated as an ideal by any set of elements. Also, no nonzero element of $\mathcal{J}$ is a finite product of irreducibles of A .*

Proof. The inclusion $\mathcal{J}^2 \subseteq \mathcal{J}$ follows from the ideal property. For $j \neq 0$, the factorization in theorem 4.2 has both factors in $\mathcal{J}$, proving the reverse inclusion.

Suppose a set $S \subseteq \mathcal{J}$ generated $\mathcal{J}$. Choose $\epsilon > 0$ strictly below every exponent in its supports. Each finite A -linear combination of elements of S has support strictly above ϵ : multiplication by an element of A only adds nonnegative exponents. But $\omega^{\epsilon/2} \in \mathcal{J}$ cannot have such a support. This is a contradiction.

Finally, if $j = p_1 \cdots p_k \neq 0$ and every p_i were irreducible, then $0 = \text{ct}(j) = \prod_i \text{ct}(p_i)$ would put some p_i in $\mathcal{J}$. That factor would be reducible by theorem 4.2. $\square$

This is an explicit failure of atomic factorization, not merely a failure of uniqueness of factorization. It leaves room for nonconstant prime elements having nonzero constant coefficient. For example, L’Innocente–Mantova prove that $\omega^{\sqrt{2}} + \omega + 1$ is prime [3, Theorem B]. That theorem is not needed in any subsequent proof here.

In particular, $\mathcal{J}^k = \mathcal{J}$ for every ordinary $k \geq 1$, so $\mathcal{J}/\mathcal{J}^2 = 0$ and the $\mathcal{J}$ -adic inverse system is constantly $A/\mathcal{J} \cong \mathbb{Z}$. Its completion is $\mathbb{Z}$, with kernel $\mathcal{J}$. These consequences appear in manuscripts 02 and 05; a zero quotient $\mathcal{J}/\mathcal{J}^2$ does not imply that $\mathcal{J}$ itself is zero.

Corollary 4.4 (Maps into a nilpotent ideal). *Let $\phi : A \rightarrow B$ be a ring homomorphism and N an ideal of B with $N^k = 0$ for some ordinary $k \geq 1$. If $\phi(\mathcal{J}) \subseteq N$, then $\phi(\mathcal{J}) = 0$.*

Proof. By corollary 4.3, $\mathcal{J}^k = \mathcal{J}$. The image under ϕ of every finite sum of k -fold products from $\mathcal{J}$ belongs to $N^k = 0$, which proves the claim. The image need not itself be an ideal of all of B . $\square$

This is manuscript 02’s nilpotent-test corollary. Its hypothesis concerns the image of the whole ideal, not merely finitely many selected elements.

Proposition 4.5 (A necessary residue condition for irreducibility). *The constant irreducibles of A are exactly the ordinary signed primes. Every nonconstant irreducible has constant coefficient 1 or -1 . The latter condition is not sufficient for irreducibility.*

Proof. A factorization of a nonzero ordinary integer has only constant factors by degree additivity, so ordinary primality decides constant irreducibility. If a nonconstant element a has $|\text{ct}(a)| > 1$, choose an ordinary prime p dividing its constant coefficient. Then $a = p(a/p)$ is a factorization into two nonunits: the second factor remains nonconstant. If $\text{ct}(a) = 0$, the factorization $a = 2(a/2)$ has the same property. This leaves only constant coefficients ± 1 . Finally, $(\omega + 1)^2$ has constant coefficient 1 and is reducible. $\square$

The factorization $j = 2(j/2)$ already proves that a nonzero $j \in \mathcal{J}$ is reducible, even in a fixed exponent workspace. The stronger content of theorem 4.2 is the existence of a common infinite monomial divisor for an entire set, with all quotients still in $\mathcal{J}$.

4.2 Clearing arbitrary set-sized families of denominators

Theorem 4.6 (Global monomial clearing). *For every set $S \subseteq \mathbf{No}$ there is $h > 0$ such that*

$$\omega^h \in \mathcal{J}, \quad \omega^h x \in \mathcal{J} \quad (x \in S).$$

Consequently $\text{Frac}(A) = \mathbf{No}$, where the equality means that every surreal is a quotient of two omnific integers with nonzero denominator.

Proof. Take the set E obtained by uniting the supports of all elements of S . Choose $h > 0$ strictly greater than every $-e$ for $e \in E$. Multiplication by ω^h shifts every exponent to $h + e > 0$. All shifted normal forms belong to $\mathcal{J}$. Applying this to $S = \{x\}$ gives $x = (\omega^h x)/\omega^h$. The reverse inclusion in the fraction field identity holds because A is a subring of $\mathbf{No}$. $\square$

Corollary 4.7 (Common multiples of every ordinary power). *Every set-sized family $(a_j)_{j \in T}$ of nonzero omnific integers has a nonzero monomial $M \in A$ such that $M/a_j^k \in \mathcal{J}$ for every $j \in T$ and every ordinary $k \geq 1$. Thus M is a common multiple of all these powers in A .*

Proof. The family of inverses a_j^{-k} is set-sized. Apply theorem 4.6 to it and take $M = \omega^h$ from that theorem. $\square$

This is manuscript 02's strengthened common-multiple statement. It does not give one element divisible by every nonzero omnific integer: the whole class of such integers is not an allowed index set.

For example, a support unbounded below in a chosen real exponent group is still a set of surreal exponents and can be shifted above zero by a new, larger surreal exponent. The theorem does not assert that the required monomial lies in the original fixed Hahn workspace.

Corollary 4.8. *Every point of $\mathbb{P}^n(\mathbf{No})$ has homogeneous coordinates in $\mathcal{J}^{n+1}$. More generally, any fixed set-sized collection of coordinate tuples can be cleared by a single positive monomial.*

Proof. Apply [theorem 4.6](#) to the set of all coordinates and multiply all tuples by the resulting nonzero scalar. $\square$

This is projective *representability*, not primitivity. In fact the coordinates in this construction always have a common nonunit divisor. Section 9 will show that some real projective directions have no primitive omnific representative at all.

4.3 Division with remainder is not a terminating Euclidean algorithm

Proposition 4.9. *For $a, b \in A$ with $b > 0$, there are unique $q, r \in A$ such that*

$$a = bq + r, \quad 0 \leq r < b.$$

Proof. Take $q = \lfloor a/b \rfloor_A$ from [theorem 2.6](#) and $r = a - bq$. The floor inequalities give the required remainder bounds. Conversely those bounds force $q \leq a/b < q+1$, so floor uniqueness determines q and then r . $\square$

Thus the familiar division step exists. Termination of repeated division is the missing ingredient: the positive order on A is not well founded. For an explicit example, start with

$$r_0 = \sqrt{2}\omega, \quad r_1 = \omega, \quad \alpha = \sqrt{2} - 1 \in (0, 1).$$

The first quotient is 1 and the first remainder is $r_2 = \alpha\omega$. Every subsequent quotient is 2, and

$$r_k = \alpha^{k-1}\omega \quad (k \geq 1).$$

Indeed, $1/\alpha = \sqrt{2} + 1$ has ordinary floor 2, and $1 - 2\alpha = \alpha^2$. Every displayed remainder is positive and infinite, so the ordinary finite Euclidean procedure never terminates.

Proposition 4.10 (An explicit pair without a gcd). *The elements $\sqrt{2}\omega$ and ω have no greatest common divisor in A , in the divisibility sense. In particular, A is not a gcd domain or a Bézout domain.*

Proof. Let $d \neq 0$ be a common divisor and put $\delta = \deg_\omega(d)$. Degree additivity gives $0 \leq \delta \leq 1$. If $\delta = 1$, both quotients $\sqrt{2}\omega/d$ and ω/d are degree-zero omnific integers, hence nonzero ordinary integers. Their ratio would make $\sqrt{2}$ rational. Therefore $\delta < 1$.

Put $\eta = (\delta + 1)/2$, so $\delta < \eta < 1$. The element ω^η is also a common divisor, since both quotients have strictly positive exponent $1 - \eta$. But ω^η cannot divide d in A , as its degree is larger than δ . Thus no proposed common divisor is divisible by every common divisor, as a gcd would be. A Bézout domain would supply a gcd for each pair, so that possibility is excluded as well. $\square$

4.4 Non-Noetherian and non-normal behavior

The strictly increasing chain

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \dots$$

already witnesses a failure of the ascending chain condition on ideals. The inclusions hold because the quotients of consecutive generators have positive exponents. They are strict because the reverse quotients have negative exponents and are not in A .

Proposition 4.11. *The full ring A is not integrally closed in its fraction field.*

Proof. Since $\mathbf{No}$ is real closed, let $r = \sqrt{\omega^2 + 1} > 0$. By [theorem 4.6](#), $r \in \text{Frac}(A)$, and it is integral over A because it satisfies the monic equation $T^2 - (\omega^2 + 1) = 0$. If $r \in A$, then

$$(r - \omega)(r + \omega) = 1.$$

Both factors would be units of A , hence would lie in $\{1, -1\}$. Their difference 2ω would then be an ordinary integer, a contradiction. Thus $r \notin A$. $\square$

Manuscript 05 gives a shorter witness to the same conclusion: $\sqrt{2} = (\sqrt{2}\omega)/\omega \in \text{Frac}(A)$ satisfies $T^2 - 2 = 0$ but is not omnific. The proof above retains the base manuscript's witness with nonconstant normal form; neither assertion depends on a factorization classification.

In particular, the full omnific ring is not a UFD or a PID. Its factorization theory cannot be replaced by an ordinary Euclidean algorithm or by an unstated gcd assumption. The modern theory in [\[3\]](#) is substantially subtler. The refinement property is different from integral closedness or atomicity; none of the preceding arguments refutes it. [Appendix B](#) records why we make no current open-status claim concerning Conway's refinement problem.

5 Equational transfer and linear Diophantine equations

5.1 Existence is exactly ordinary existence

For a finite family of polynomials $F_1, \dots, F_r$ over a ring R , write $V_R(F_1, \dots, F_r)$ for its simultaneous zero set in R^n .

Theorem 5.1 (Retraction theorem for equations). *Let $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$. Coordinatewise constant coefficient gives a retraction*

$$V_A(F_1, \dots, F_r) \longrightarrow V_{\mathbb{Z}}(F_1, \dots, F_r).$$

In particular,

$$V_A(F_1, \dots, F_r) \neq \emptyset \iff V_{\mathbb{Z}}(F_1, \dots, F_r) \neq \emptyset. \quad (5.1)$$

The same equivalence holds for positive existential formulas in the ring language, evaluated at ordinary integer parameters.

Proof. For every polynomial F over $\mathbb{Z}$ and $\mathbf{x} \in A^n$,

$$\text{ct}(F(\mathbf{x})) = F(\text{ct}(x_1), \dots, \text{ct}(x_n)),$$

because ct is a ring homomorphism fixing all coefficients. Thus the constant tuple of an omnific solution is an integer solution. Conversely, an integer solution is already an omnific solution, and applying ct to it does nothing.

A positive existential formula is built from equalities using finite conjunction, finite disjunction, and existential quantification. Ring homomorphisms preserve such formulas, as follows directly by induction on their formation; witnesses are mapped coordinatewise. Apply this first to ct , and then to the inclusion of $\mathbb{Z}$ in A . $\square$

The theorem answers a question about existence, not cardinalities or complete solution sets. For $Y = X^2$, the fiber of the retraction above $(0, 0)$ already contains (t, t^2) for every $t \in \mathcal{J}$.

Corollary 5.2 (Hilbert’s tenth problem). *There is no algorithm deciding, from an ordinary integer polynomial, whether it has a zero in A . The yes-instances are exactly the ordinary-integer yes-instances of Hilbert’s tenth problem, and form a computably enumerable complete set under the usual computable reductions.*

Proof. Equation (5.1) identifies the two decision problems on exactly the same finite inputs. The undecidability and completeness of the ordinary problem are the Davis–Putnam–Robinson–Matiyasevich (MRDP) theorem and its standard consequences [7]. Enumerating ordinary integer tuples semidecides the omnific yes-instances as well; no enumeration of the proper class A is involved. $\square$

This corollary concerns *ordinary integer coefficients*. A decision problem allowing arbitrary omnific coefficients requires an explicit finite representation scheme, and is a different question.

5.2 Why signs and nonvanishing do not transfer

Consider

$$(X + 1)^2 - 2(Y + 1)^2 = 0.$$

It has no solution in $\mathbb{N}^2$, since $X + 1$ and $Y + 1$ would be positive integers with rational ratio $\sqrt{2}$. However,

$$X = \sqrt{2}\omega - 1, \quad Y = \omega - 1$$

is a solution with both coordinates positive in A . Its constant tuple is $(-1, -1)$, an ordinary solution of the equation without positivity constraints.

Proposition 5.3. *With ordinary integer parameters, none of the subsets*

$$A \setminus \{0\}, \quad A^{>0}, \quad A^{\geq 0}$$

is definable by a positive existential ring formula over A .

Proof. Every such definable subset must be stable under ct by the homomorphism argument in [theorem 5.1](#). But ω is nonzero and positive while $\text{ct}(\omega) = 0$. Also $\omega - 1 > 0$ whereas $\text{ct}(\omega - 1) = -1 < 0$. $\square$

Thus “nonzero” cannot be harmlessly added to an equational transfer statement. Encoding nonvanishing by $XY = 1$ would restrict X to $\{1, -1\}$, not to all nonzero omnific integers.

5.3 A complete solution criterion for linear systems

Theorem 5.4 (Smith normal form over the omnific integers). *Let $M \in \mathbb{Z}^{m \times n}$ and $\mathbf{b} \in A^m$. Choose unimodular integer matrices U, V with*

$$UMV = D = \text{diag}(d_1, \dots, d_r, 0), \quad d_i > 0,$$

in the rectangular Smith normal form, and put $\mathbf{c} = U\mathbf{b}$. Then $M\mathbf{x} = \mathbf{b}$ is solvable in A^n if and only if

$$d_i \mid \text{ct}(c_i) \quad (1 \leq i \leq r), \quad c_i = 0 \quad (r < i \leq m). \quad (5.2)$$

All solutions are $\mathbf{x} = V\mathbf{y}$, where

$$y_i = c_i/d_i \quad (1 \leq i \leq r), \quad y_{r+1}, \dots, y_n \in A \text{ are arbitrary.}$$

Proof. Unimodular integer matrices and their inverses act on omnific tuples. The substitution $\mathbf{x} = V\mathbf{y}$ transforms the system into $D\mathbf{y} = \mathbf{c}$. For each nonzero diagonal entry, division is possible precisely under [theorem 3.1](#). Zero rows require the full equation $c_i = 0$, not merely its constant coefficient. The remaining coordinates are unconstrained. $\square$

For ordinary $\mathbf{b}$, a useful equivalent description is

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad M\mathbf{a} = \mathbf{b}, \quad \mathbf{u} \in \mathcal{J}^n, \quad M\mathbf{u} = 0.$$

If M has full column rank, all solutions are ordinary. For a concrete one-equation example, every solution to $2x + 3y = 1$ is

$$x = -1 + 3t, \quad y = 1 - 2t, \quad t \in A.$$

The same formulas as over $\mathbb{Z}$ work, but the parameter class is larger.

6 Rigidity of constant-product and norm equations

6.1 The constant-product principle

Lemma 6.1. *If $b_1, \dots, b_s \in \mathcal{B}_{\mathbb{C}}$ and $b_1 \cdots b_s = c \in \mathbb{C}^\times$, then every b_i belongs to $\mathbb{C}^\times$.*

Proof. Each b_i has inverse $c^{-1} \prod_{j \neq i} b_j$ in $\mathcal{B}_{\mathbb{C}}$. Apply [proposition 2.5](#). Equivalently, their nonnegative degrees have sum zero, so every degree is zero. $\square$

The importance of $\mathcal{B}_{\mathbb{C}}$ is that linear factors may have irrational or complex coefficients. They need not themselves be omnific integers. They do, however, have no negative exponents, so the unit argument applies.

Theorem 6.2 (Decomposable polynomial fibers). *Let*

$$F(\mathbf{X}) = \lambda \prod_{j=1}^s L_j(\mathbf{X}), \quad \lambda \in \mathbb{C}^\times,$$

where the L_j are affine-linear polynomials over $\mathbb{C}$. Suppose the matrix of their linear parts has column rank n . If $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$ and $F(\mathbf{x}) = c \in \mathbb{C}^\times$, then $\mathbf{x} \in \mathbb{C}^n$. In particular, any such point in A^n lies in $\mathbb{Z}^n$.

Proof. By [lemma 6.1](#), every $L_j(\mathbf{x})$ is constant. Select n rows whose linear coefficient matrix is invertible over $\mathbb{C}$. Subtract the constant terms and solve this ordinary linear system. All coordinates of $\mathbf{x}$ are complex constants. Its intersection with A^n is $\mathbb{Z}^n$. $\square$

The rank hypothesis is essential. The equation $X^d = 1$ in two variables forces X to be constant but says nothing about Y .

6.2 Binary forms, Pell equations, and power differences

Theorem 6.3 (Binary-form rigidity). *Let $F \in \mathbb{Z}[X, Y]$ be a nonzero homogeneous binary form having at least two distinct projective linear factors over $\mathbb{C}$. For each $c \in \mathbb{Z} \setminus \{0\}$,*

$$\{(x, y) \in A^2 : F(x, y) = c\} = \{(x, y) \in \mathbb{Z}^2 : F(x, y) = c\}.$$

Proof. Factor F into linear forms over $\mathbb{C}$, keeping their multiplicities. Two distinct projective factors have linearly independent coefficient vectors. The rank in [theorem 6.2](#) is therefore two. $\square$

This theorem includes the nonzero fibers of every irreducible binary form of degree at least two over $\mathbb{Q}$. For irreducible forms of degree at least three these are the usual Thue equations; the conclusion here is about the absence of additional omnific solutions, not a new proof of ordinary finiteness or an algorithm for the ordinary solutions.

Corollary 6.4 (Pell rigidity). *For ordinary integers $D \neq 0$ and $c \neq 0$, every solution in A^2 of*

$$x^2 - Dy^2 = c$$

is an ordinary integer solution. In particular,

$$\{(u, v) \in A^2 : u^2 - 2v^2 = 1\} = \{(u, v) \in \mathbb{Z}^2 : u^2 - 2v^2 = 1\}.$$

Proof. Over $\mathbb{C}$, the two factors $X - \sqrt{D}Y$ and $X + \sqrt{D}Y$ are distinct because $D \neq 0$. Apply [theorem 6.3](#). $\square$

The classical Pell equation has ordinary solutions of arbitrarily large finite size. It nevertheless has no single infinite omnific solution. This distinction is a first warning against treating A as an elementary extension of ordinary integer arithmetic.

In the full surreal field, the contrast is visible without solving a recurrence. For $D > 0$, $c \neq 0$, and $s \in \mathbf{No}^\times$,

$$x = \frac{s + c/s}{2}, \quad y = \frac{s - c/s}{2\sqrt{D}}$$

satisfy $x^2 - Dy^2 = c$. Taking $s = \omega$ produces negative-exponent terms, so neither coordinate is an omnific integer. Discarding those terms changes the norm from c to zero. A surreal solution cannot be turned into an omnific solution by unverified truncation.

Proposition 6.5 (The zero Pell fiber). *If D is a positive nonsquare ordinary integer, all solutions of $x^2 - Dy^2 = 0$ in A^2 are exactly*

$$(x, y) = (\sqrt{D}t, t) \quad \text{or} \quad (-\sqrt{D}t, t), \quad t \in \mathcal{J}.$$

Proof. Factor in $\mathbf{No}$ to obtain $x = \pm\sqrt{D}y$. Write $y = m + j$ with $m \in \mathbb{Z}$ and $j \in \mathcal{J}$. The constant coefficient of x is $\pm\sqrt{D}m$, which is an integer only if $m = 0$. Conversely, multiplication of $\mathcal{J}$ by a real scalar preserves $\mathcal{J}$, so all displayed pairs belong to A^2 and solve the equation. $\square$

Thus changing the right side from a nonzero constant to zero changes the answer completely.

Corollary 6.6 (Power-difference rigidity). *Let $m, n \geq 2$ be ordinary integers with $d = \gcd(m, n) > 1$, and let $0 \neq c \in \mathbb{Z}$. Every omnific solution of $x^m - y^n = c$ is ordinary.*

Proof. Set $X = x^{m/d}$ and $Y = y^{n/d}$. The binary form $X^d - Y^d$ has at least two distinct complex factors, so X, Y are ordinary by [theorem 6.3](#). If a positive ordinary power of an element of A is an ordinary constant, that element is ordinary: for a nonzero element this follows from degree additivity, and for a zero power value the integral domain property suffices. $\square$

No conclusion for coprime exponents is being smuggled into this statement. The argument does not classify equations such as $x^2 - y^3 = 1$.

6.3 Number-field norm equations

Theorem 6.7 (Norm-form rigidity). *Let $K/\mathbb{Q}$ be a number field of degree d , let $e_1, \dots, e_d$ be a $\mathbb{Q}$ -basis, and let*

$$N(\mathbf{X}) = N_{K/\mathbb{Q}} \left(\sum_{j=1}^d e_j X_j \right).$$

For every nonzero rational constant c , every solution $\mathbf{x} \in A^d$ of $N(\mathbf{x}) = c$ has all coordinates in $\mathbb{Z}$. In particular, with an integral basis this applies to the usual integer-coefficient norm equations.

Proof. The d embeddings $\sigma_1, \dots, \sigma_d : K \rightarrow \mathbb{C}$ give

$$N(\mathbf{X}) = \prod_{i=1}^d \left(\sum_{j=1}^d \sigma_i(e_j) X_j \right).$$

The embedding matrix $(\sigma_i(e_j))$ is invertible. To verify this, choose a primitive element θ of the separable extension $K/\mathbb{Q}$. For the power basis $1, \theta, \dots, \theta^{d-1}$ the matrix is Vandermonde in the distinct conjugates of θ , so has nonzero determinant. A basis change over $\mathbb{Q}$ preserves invertibility. The result now follows from [theorem 6.2](#). $\square$

For example, consider the norm form for $K = \mathbb{Q}(\sqrt[3]{2})$:

$$N(X + Y\sqrt[3]{2} + Z\sqrt[3]{4}) = X^3 + 2Y^3 + 4Z^3 - 6XYZ.$$

Every omnific solution of

$$X^3 + 2Y^3 + 4Z^3 - 6XYZ = 1$$

is an ordinary integer triple. This is a genuinely three-variable rigidity result, unlike the indefinite quadratic forms in the next section.

6.4 An algebraic-geometric formulation

Let V be an affine complex variety whose coordinate ring $\mathbb{C}[V]$ is generated, as a $\mathbb{C}$ -algebra, by units. Every $\mathcal{B}_{\mathbb{C}}$ -point of V is then a constant $\mathbb{C}$ -point. Indeed, a $\mathbb{C}$ -algebra map $\mathbb{C}[V] \rightarrow \mathcal{B}_{\mathbb{C}}$ sends each generating unit to an element of $\mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}$, so its entire image lies in $\mathbb{C}$. This includes closed subvarieties of an algebraic torus, in their very-affine coordinate presentation.

The statement supplies a conceptual explanation of the norm argument: a nonzero product relation forces its factors into a constant torus. It is not a claim that every affine variety is constant over $\mathcal{B}_{\mathbb{C}}$. The affine line, for example, has the full class $\mathcal{B}_{\mathbb{C}}$ of points.

7 A complete dichotomy for nonzero quadratic levels

Binary Pell rigidity might suggest that a nonzero constant quadratic level never contains infinite points. This is false as soon as there is enough indefinite dimension. The following result gives a precise classification.

Theorem 7.1 (Quadratic-level dichotomy). *Let $n \geq 1$, let $q \in \mathbb{Z}[X_1, \dots, X_n]$ be a homogeneous quadratic form, and let $c \in \mathbb{Z} \setminus \{0\}$. Write*

$$V_A = \{\mathbf{x} \in A^n : q(\mathbf{x}) = c\}, \quad V_{\mathbb{Z}} = \{\mathbf{a} \in \mathbb{Z}^n : q(\mathbf{a}) = c\}.$$

If $V_{\mathbb{Z}}$ is empty, so is V_A . Otherwise the following alternatives are exhaustive.

- (i) *If q is nondegenerate and is either definite or of dimension at most two, then $V_A = V_{\mathbb{Z}}$.*
- (ii) *If q is degenerate, every $\mathbf{a} \in V_{\mathbb{Z}}$ lies on an injective affine-linear family $\mathbf{a} + t\mathbf{v}$, $t \in \mathcal{J}$, in V_A , with every $t \neq 0$ giving an infinite point.*
- (iii) *If q is nondegenerate, indefinite, and $n \geq 3$, every $\mathbf{a} \in V_{\mathbb{Z}}$ lies on an injective quadratic polynomial family $\mathbf{a} + t\mathbf{v} + t^2\mathbf{w}$, $t \in \mathcal{J}$, in V_A , again with every $t \neq 0$ giving an infinite point.*

In alternatives (ii) and (iii), the constant tuple of every point in the family is $\mathbf{a}$.

We first prove the geometric construction required for the indefinite case. All vectors and bilinear forms in its statement are ordinary real objects; only the parameter is later allowed to be surreal.

Lemma 7.2 (A polynomial quadratic isometry). *Let B be a symmetric bilinear form over $\mathbb{R}$, with $q(x) = B(x, x)$. Suppose e, u satisfy*

$$q(e) = 0, \quad B(e, u) = 0, \quad k = q(u) \neq 0.$$

For a scalar parameter t define

$$T_t(x) = x + t(B(x, e)u - B(x, u)e) - \frac{k}{2}t^2B(x, e)e. \quad (7.1)$$

Over every field extension of $\mathbb{R}$,

$$q(T_t(x)) = q(x), \quad T_s T_t = T_{s+t}, \quad T_t^{-1} = T_{-t}.$$

If $\alpha = B(a, e) \neq 0$, the curve $t \mapsto T_t(a)$ is injective.

Proof. Put $\alpha = B(x, e)$ and $\beta = B(x, u)$. In (7.1), the added vector is

$$t\alpha u - \left(t\beta + \frac{k}{2}t^2\alpha\right)e.$$

Its scalar product with x , multiplied by two, is $-kt^2\alpha^2$, because the two linear terms cancel. Its own quadratic value is $kt^2\alpha^2$, since e is isotropic and orthogonal to u . Their sum is zero, proving preservation of q .

For the group law, set $N(x) = B(x, e)u - B(x, u)e$. Then

$$N^2(x) = -kB(x, e)e, \quad N^3 = 0, \quad T_t = \text{id} + tN + \frac{t^2}{2}N^2.$$

Multiplying these finite polynomials in N gives $T_s T_t = T_{s+t}$, and hence the inverse formula.

Finally, for $\alpha = B(a, e) \neq 0$ and $\beta = B(a, u)$,

$$T_t(a) = a + tv + t^2 w, \quad v = \alpha u - \beta e, \quad w = -\frac{k\alpha}{2}e.$$

The real vectors v, w are linearly independent: w is a nonzero multiple of e , whereas v has nonzero u -component, and u, e are independent. Their independence persists over a field extension because some 2×2 real minor is nonzero. If $T_s(a) = T_t(a)$ and $s \neq t$, then $v + (s + t)w = 0$, contradicting that independence. $\square$

Lemma 7.3. *If q is a nondegenerate indefinite real quadratic form of dimension at least three and $q(a) \neq 0$, there exist e, u as in [lemma 7.2](#) with $B(a, e) \neq 0$.*

Proof. The real isotropic vectors span the entire space. To see this, use a real basis in which q has positive unit vectors p_i and negative unit vectors n_j . Both types exist by indefiniteness, and each $p_i + n_j$ and $p_i - n_j$ is isotropic. Taking sums and differences recovers every basis direction. Since the nonzero functional $B(a, -)$ cannot vanish on a spanning set, some isotropic e has $B(a, e) \neq 0$.

Nondegeneracy gives f_0 with $B(e, f_0) = 1$. Replacing it by $f = f_0 - \frac{1}{2}q(f_0)e$ makes $q(f) = 0$ while retaining $B(e, f) = 1$. The plane spanned by e, f is nondegenerate. Its orthogonal complement is nondegenerate and has dimension $n - 2 \geq 1$, so it contains u with $q(u) \neq 0$. This u is orthogonal to e , as required. $\square$

Proof of [theorem 7.1](#). The empty-set assertion is [theorem 5.1](#).

Suppose first that q is definite and $\mathbf{x} \in V_A$ has an infinite coordinate. Let $\gamma > 0$ be the largest of the leading exponents of its nonzero coordinates. Let r_i be the coefficient of ω^γ in x_i , using zero when that exponent is absent. The real vector $\mathbf{r}$ is nonzero. The coefficient of $\omega^{2\gamma}$ in $q(\mathbf{x})$ is exactly $q(\mathbf{r})$, which is nonzero by definiteness. This contradicts $q(\mathbf{x}) = c$. Thus every coordinate is finite and hence an ordinary integer. In dimension two, nondegeneracy means that the binary quadratic has two distinct linear factors over $\mathbb{C}$, so [theorem 6.3](#) applies. A nondegenerate one-dimensional form is definite. This proves (i).

For (ii), choose a nonzero real vector $v \in \text{rad}(B)$. For every t , $q(a + tv) = q(a) = c$. If $t \in \mathcal{J}$, the coordinates of tv belong to $\mathcal{J}$, so $a + tv \in A^n$ with constant tuple a . The map is injective because $v \neq 0$. For $t \neq 0$, some coordinate has positive degree.

For (iii), choose e, u by [lemma 7.3](#) and use [lemma 7.2](#). Each coordinate of $T_t(a)$ is an integer constant plus a real multiple of t plus a real multiple of t^2 . Thus it belongs to A and has constant coefficient a_i for $t \in \mathcal{J}$. Injectivity has already been proved. If $t \neq 0$, its leading degree is positive. In a coordinate where $w_i \neq 0$, the quadratic term has degree $2 \deg_\omega(t)$, strictly larger than the linear term's degree, and cannot cancel. Therefore the point is infinite. $\square$

7.1 An explicit infinite hyperboloid family

For positive ordinary integers D, E and $t \in \mathcal{J}$, set

$$x = 1 + \frac{E}{2}t^2, \quad y = \frac{E}{2\sqrt{D}}t^2, \quad z = t. \tag{7.2}$$

All three coordinates are omnific integers, and a direct expansion gives

$$x^2 - Dy^2 - Ez^2 = 1.$$

Indeed, the two quartic terms cancel, as do the two quadratic terms. For $t \neq 0$ every coordinate is infinite. Nevertheless the tuple is unimodular, since

$$1 = x - \left(\frac{E}{2}t\right)z, \quad \frac{E}{2}t \in A. \quad (7.3)$$

This illustrates a particularly sharp contrast:

$$x^2 - Dy^2 = 1 \text{ has only ordinary points,} \quad x^2 - Dy^2 - Ez^2 = 1 \text{ has infinite unimodular points.}$$

There is no appeal to convergence, analytic continuation, or infinite iteration in this construction; every expression is a finite polynomial.

7.2 What the classification does and does not say

The classification concerns the presence of genuinely new points on a *represented nonzero level*. It does not decide whether the integer level is represented in the first place. Nor does it classify all infinite points: a displayed curve is a rigorously constructed subset of the full fiber.

For zero levels, homogeneity gives a different mechanism. An indefinite real quadratic form has real isotropic directions, and multiplying any such direction by an element of $\mathcal{J}$ produces omnific points with zero constant tuple. Such points can exist even when there is no nonzero ordinary integer isotropic vector.

8 Defining the ordinary integers by one quartic equation

The retraction ct detects ordinary coefficients but is not a symbol in the ring language. We now show that the embedded copy of $\mathbb{Z}$ can be detected by a single explicit polynomial equation with existential auxiliary variables.

8.1 Pell solutions as finite bounds

Define ordinary positive Pell solutions by

$$u_k + v_k\sqrt{2} = (3 + 2\sqrt{2})^k, \quad k \geq 0. \quad (8.1)$$

Then $u_k, v_k \in \mathbb{Z}$, $u_k^2 - 2v_k^2 = 1$, and u_k tends to infinity in the ordinary real sense. The inverse of $3 + 2\sqrt{2}$ is $3 - 2\sqrt{2} \in (0, 1)$, so

$$u_k = \frac{(3 + 2\sqrt{2})^k + (3 - 2\sqrt{2})^k}{2}.$$

The recurrence

$$u_0 = 1, \quad u_1 = 3, \quad u_{k+2} = 6u_{k+1} - u_k$$

implies

$$u_{2j} \equiv 1 \pmod{8}, \quad u_{2j+1} \equiv 3 \pmod{8}. \quad (8.2)$$

Both parity subsequences are unbounded.

We use the classical Legendre–Gauss three-square theorem: a nonnegative ordinary integer N is a sum of three integer squares if and only if it is not of the form $4^a(8b + 7)$, with $a, b \geq 0$. This theorem is imported, not reproved here; see the mathematical exposition in [8] and the formal development in [9].

Lemma 8.1 (A three-square gap below a Pell coordinate). *For every ordinary nonnegative integer S , there are ordinary integers u, v, w_1, w_2, w_3 such that*

$$u^2 - 2v^2 = 1, \quad u = S + w_1^2 + w_2^2 + w_3^2.$$

Proof. Choose the residue of u modulo 8 according to the following table. Each choice is realized by an unbounded subsequence in (8.1).

$S \bmod 8$	0	1	2	3	4	5	6	7
chosen $u \bmod 8$	1	3	3	1	1	3	1	1
$(u - S) \bmod 8$	1	2	1	6	5	6	3	2

Take a sufficiently large u in that subsequence so that $N = u - S > 0$. Its residue is one of 1, 2, 3, 5, 6. Hence it is neither divisible by 4 nor congruent to 7 modulo 8, so it cannot be $4^a(8b + 7)$. The three-square theorem supplies the w_j . The corresponding Pell coordinate supplies v . $\square$

8.2 The uniform five-auxiliary-variable guard

For each ordinary $n \geq 1$, define the integer polynomial

$$G_n(\mathbf{X}, U, V, \mathbf{W}) = (U^2 - 2V^2 - 1)^2 + \left(U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2. \quad (8.3)$$

It has total degree four and $n + 5$ variables.

Theorem 8.2 (Uniform quartic integrality guard). *For every $\mathbf{x} \in A^n$,*

$$\mathbf{x} \in \mathbb{Z}^n \iff \exists u, v, w_1, w_2, w_3 \in A \quad G_n(\mathbf{x}, u, v, w_1, w_2, w_3) = 0. \quad (8.4)$$

In fact, every coordinate of every zero of G_n in A^{n+5} is an ordinary integer. The number of auxiliary variables is five, independently of n .

Proof. Suppose $G_n = 0$ in A . A sum of two squares in an ordered field can vanish only when both squares vanish. Thus

$$u^2 - 2v^2 = 1, \quad u = \sum_i x_i^2 + w_1^2 + w_2^2 + w_3^2. \quad (8.5)$$

By Pell rigidity, $u, v \in \mathbb{Z}$. The second equation gives $u \geq 0$ and bounds the square of each x_i and w_j by the ordinary number u . Consequently every x_i, w_j is finite. By [proposition 2.5](#), each is an ordinary integer.

Conversely, suppose $\mathbf{x} \in \mathbb{Z}^n$ and put $S = \sum_i x_i^2 \in \mathbb{N}$. Choose an ordinary Pell solution and three squares from [lemma 8.1](#). They satisfy both equations in (8.5), and hence make G_n vanish. $\square$

The theorem is uniform in tuple length, but does not claim that five auxiliary variables are optimal. A slightly simpler version using four square variables instead of three follows from Lagrange's four-square theorem and arbitrary sufficiently large Pell coordinates; the modular argument above removes one auxiliary variable.

For $x = 1$, one witness is $u = 3$, $v = 2$, and $(w_1, w_2, w_3) = (1, 1, 0)$. For $x = 2$, a witness is $u = 17$, $v = 12$, and $(w_1, w_2, w_3) = (3, 2, 0)$. The assertions for all integers rely on [lemma 8.1](#), not on these examples or on a finite computation.

8.3 Definability consequences and a logical distinction

Let $\text{Int}(x)$ abbreviate the existential formula obtained from G_1 . Then $\text{Int}(A) = \mathbb{Z}$. Thus ordinary integer arithmetic is present not only as the quotient $A/\mathcal{J}$ but as an existentially definable subring of A itself.

There is no conflict with [proposition 5.3](#). The copy of $\mathbb{Z}$ is stable under ct , whereas the full positive cone and the nonzero locus are not. Nor does this make A an elementary extension of $\mathbb{Z}$: the formula $\text{Int}(x)$ holds for every element of $\mathbb{Z}$ but fails for ω in A .

Corollary 8.3. *The ideal $\mathcal{J}$ and the graph of ct are first-order definable in the pure ring structure of A , with no nonstandard parameters. Explicitly,*

$$\begin{aligned} j \in \mathcal{J} &\iff \forall n ((\text{Int}(n) \wedge n \neq 0) \implies \exists y (j = ny)), \\ m = \text{ct}(x) &\iff \text{Int}(m) \wedge (x - m \in \mathcal{J}). \end{aligned}$$

Proof. The first formula restricts the divisor variable to the ordinary nonzero integers, then applies (3.2). The second expresses the unique decomposition $x = m + j$ with $m \in \mathbb{Z}$, $j \in \mathcal{J}$. $\square$

The displayed definition of $\mathcal{J}$ uses a universal quantifier and an implication. We do not claim that it is an existential or Diophantine definition of $\mathcal{J}$.

There is also a concrete failure of a familiar arithmetic sentence. The positive omnific integer $\omega - 1$ is not a sum of any fixed finite number of omnific squares. Applying ct to such a proposed identity would give

$$-1 = m_1^2 + \cdots + m_r^2 \quad (m_i \in \mathbb{Z}),$$

which is impossible. In particular, the ordinary assertion that every nonnegative integer is a sum of four squares fails for A 's full nonnegative cone. This is compatible with the classical fact that integer parts of real closed fields satisfy Open Induction [6, 3]; Open Induction is not full ordinary arithmetic.

8.4 A reusable abstract principle

The proof of [theorem 8.2](#) needs less than all surreal numbers. Let D be any discretely ordered ring containing $\mathbb{Z}$, and suppose that every solution in D of $U^2 - 2V^2 = 1$ is ordinary. Every element of D bounded by an ordinary integer is itself ordinary: discreteness and a finite comparison with the integers in the bounding interval prove this. The identical proof therefore shows that G_n defines $\mathbb{Z}^n$ in D^n . Our contribution is an explicit guard under a transparent Pell-rigidity hypothesis, together with a proof that the omnific rings under consideration satisfy that hypothesis.

9 Infinite families, primitivity, and projective points

9.1 Polynomial curves through integer points

Proposition 9.1 (Finite polynomial lifting). *Let $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$, and let $P_1(T), \dots, P_n(T) \in \mathbb{R}[T]$ satisfy*

$$F_j(P_1(T), \dots, P_n(T)) = 0 \quad (1 \leq j \leq r), \quad P_i(0) \in \mathbb{Z} \quad (1 \leq i \leq n).$$

Then $\mathbf{P}(t) \in A^n$ solves the system for every $t \in \mathcal{J}$, with constant tuple $\mathbf{P}(0)$. If some P_i is nonconstant, every $t \in \mathcal{J} \setminus \{0\}$ produces an infinite point.

Proof. Every nonconstant term of $P_i(t)$ lies in $\mathcal{J}$, so the constant coefficient is exactly the integer $P_i(0)$. Polynomial identities hold under substitution into any field extension. If P_i has degree $d > 0$ and $\deg_\omega(t) = \gamma > 0$, its top term has degree $d\gamma$, strictly larger than every other term's degree, so cannot cancel. $\square$

The finiteness of the polynomials matters. Evaluating an infinite real power series at an infinite omnific argument is not licensed by this proposition. For instance, the Taylor expansion of $\sqrt{1+T}$ at zero is not an admissible way to assert that $\sqrt{1+\omega}$ lies in A .

9.2 Infinite solutions of homogeneous equations

Theorem 9.2 (Homogeneous cone criterion). *Let $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$ be homogeneous polynomials of positive degrees. Their common zero locus has a nonzero point in A^n if and only if it has a nonzero point in $\mathbb{R}^n$.*

Proof. A real nonzero common zero $\mathbf{a}$ gives the omnific common zero $\omega\mathbf{a} \in \mathcal{J}^n$ by homogeneity.

Conversely, take a nonzero common zero $\mathbf{x} \in A^n$. If every coordinate is constant, it is already a real point. Otherwise let $\gamma > 0$ be the maximum of the coordinates' leading degrees and let $r_i = \lfloor \omega^\gamma \rfloor x_i$. The vector $\mathbf{r} \in \mathbb{R}^n$ is nonzero. For a homogeneous F_j of degree d_j , the coefficient at $\omega^{d_j\gamma}$ in $F_j(\mathbf{x})$ is $F_j(\mathbf{r})$: a product can attain that exponent only by taking the exponent γ in every factor. Since $F_j(\mathbf{x}) = 0$, this coefficient vanishes. Thus $\mathbf{r}$ is a nonzero real common zero. $\square$

For example, for every ordinary integer $n \geq 2$,

$$x = t, \quad y = t, \quad z = 2^{1/n}t, \quad 0 < t \in \mathcal{J} \quad (9.1)$$

gives a positive infinite solution of $x^n + y^n = z^n$. The irrational real coefficient of t is permitted by the omnific definition. All three constant coefficients are zero, so these families carry no nonzero ordinary integer solution under the retraction.

All coordinates in (9.1) are divisible by 2 in A , since they belong to $\mathcal{J}$. By [theorem 4.2](#), they also have a common infinite monomial divisor. Hence these are not primitive solutions. One must not cite them as resolving a primitive Fermat problem.

9.3 Primitive versus unimodular tuples

Definition 9.3. A nonzero tuple over A is *primitive* if it has no common nonunit divisor. It is *unimodular* if its coordinates generate the unit ideal, equivalently if an A -linear combination of them is 1.

Unimodularity implies primitivity. The converse is not being assumed in this non-PID ring. We use explicit identities whenever a tuple is claimed to be unimodular.

Proposition 9.4 (Infinite unimodular Pythagorean triples). *For every $t \in \mathcal{J}$, put*

$$x = t^2 - 1, \quad y = 2t, \quad z = t^2 + 1.$$

Then $x^2 + y^2 = z^2$ and the tuple is unimodular. For $t \neq 0$ it is a genuinely infinite triple.

Proof. The identity is the ordinary polynomial identity $(t^2 - 1)^2 + 4t^2 = (t^2 + 1)^2$. Put $a = t^2/2 \in \mathcal{J} \subseteq A$. Then

$$az - (a + 1)x = \frac{t^2}{2}(z - x) - x = t^2 - (t^2 - 1) = 1. \quad (9.2)$$

Thus even the pair (x, z) generates the unit ideal. Positive leading degree gives the assertion about infinite size. $\square$

The constant tuple is $(-1, 0, 1)$, not the zero tuple. That is exactly where the common-divisor obstruction for $\mathcal{J}^3$ ceases to apply.

9.4 Which real projective directions have primitive representatives?

A point $[r_0 : \cdots : r_n] \in \mathbb{P}^n(\mathbb{R})$ is *rational* if it has homogeneous coordinates in $\mathbb{Q}$, equivalently a nonzero integer coordinate tuple.

Theorem 9.5 (Rationality of primitive constant directions). *For a real projective point $p = [r_0 : \cdots : r_n]$, the following are equivalent:*

- (i) p is rational;
- (ii) p has a unimodular representative in A^{n+1} ;
- (iii) p has a primitive representative in A^{n+1} .

Proof. For (i) $\Rightarrow$ (ii), choose an integer representative whose ordinary integer gcd is one. The usual integer Bézout identity makes the tuple unimodular over A as well. Implication (ii) $\Rightarrow$ (iii) follows from the definition.

For (iii) $\Rightarrow$ (i), write a primitive representative as $z_i = \lambda r_i$ with $\lambda \in \mathbf{No}^\times$. Choose $r_j \neq 0$. Since $z_j \in A \subseteq \mathcal{B}_{\mathbb{R}}$ and division by the nonzero real scalar r_j preserves $\mathcal{B}_{\mathbb{R}}$, we have $\lambda \in \mathcal{B}_{\mathbb{R}}$. Put $c = \text{ct}(\lambda) \in \mathbb{R}$. Then

$$\text{ct}(z_i) = cr_i \in \mathbb{Z} \quad (0 \leq i \leq n).$$

If $c = 0$, every z_i belongs to $\mathcal{J}$, so 2 divides every coordinate in A , contradicting primitivity. Hence $c \neq 0$, and $(cr_0, \dots, cr_n)$ is a nonzero integer representative of p . $\square$

For example, $[\sqrt{2} : 1]$ has omnific representatives such as $(\sqrt{2}\omega, \omega)$, but has no primitive omnific representative. In contrast, each rational real direction has an ordinary coprime integer representative. The theorem concerns real projective points; it is not a classification of all points of $\mathbb{P}^n(\mathbf{No})$.

10 Obstructions from leading terms and bounded geometry

10.1 Initial equations are necessary, not sufficient

Let $F(\mathbf{X}) = \sum_{\alpha} a_{\alpha} \mathbf{X}^{\alpha}$ be a nonzero polynomial over A , and suppose all coordinates of a proposed solution $\mathbf{x} \in A^n$ are nonzero. For nonzero coefficients put

$$\lambda_{\alpha} = \deg_{\omega}(a_{\alpha}), \quad c_{\alpha} = \text{lc}(a_{\alpha}), \quad \gamma_i = \deg_{\omega}(x_i), \quad r_i = \text{lc}(x_i).$$

The largest possible exponent in $F(\mathbf{x})$ is

$$M = \max_{\alpha} \left(\lambda_{\alpha} + \sum_i \alpha_i \gamma_i \right).$$

If $F(\mathbf{x}) = 0$, at least two monomials must attain M , and

$$\sum_{\lambda_{\alpha} + \sum_i \alpha_i \gamma_i = M} c_{\alpha} \prod_i r_i^{\alpha_i} = 0. \quad (10.1)$$

This follows by extracting the coefficient at M . Variables assigned zero can first be substituted out of the polynomial.

For standard coefficients this is the usual leading-weight obstruction, now with surreal weights. It must be used together with the constant-coefficient equations. Neither set of necessary conditions is sufficient in general.

For example, $Y^2 = \omega^2 + 1$ passes the constant test $\text{ct}(Y)^2 = 1$ and the leading test $\gamma_Y = 1$, $r_Y = \pm 1$. Nevertheless, there is no solution in A , by the unit argument in [proposition 4.11](#). The missing information resides in the full tail of a potential root.

10.2 No real point at infinity implies rigidity

Theorem 10.1 (Leading-homogeneous obstruction). *Let $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$ be non-constant polynomials, and let H_j be the top homogeneous part of F_j . If the H_j have no common nonzero real zero, then every common zero of the F_j in A^n lies in $\mathbb{Z}^n$.*

Proof. Suppose a common zero has an infinite coordinate. Choose its maximum coordinate degree $\gamma > 0$ and nonzero leading vector $\mathbf{r}$ as in [theorem 9.2](#). If $d_j = \deg F_j$, the coefficient at $d_j \gamma$ in $F_j(\mathbf{x})$ is $H_j(\mathbf{r})$. Every term of lower ordinary total degree has exponent strictly less than $d_j \gamma$, and therefore cannot affect that coefficient. Thus every $H_j(\mathbf{r})$ is zero, contrary to the hypothesis. All coordinates are consequently finite and hence ordinary integers. $\square$

The hypothesis can be viewed as a sufficient absence-of-real-directions- at-infinity condition for the chosen system. Its converse is false. A Pell equation has real directions at infinity when $D > 0$, but its nonzero fibers are rigid by factorization. Initial-form obstructions and constant-product obstructions detect different mechanisms.

10.3 Positive bounds and orthogonal matrices

Proposition 10.2. *If $a_1, \dots, a_n$ are positive ordinary real numbers, $x_1, \dots, x_n \in A$, and*

$$\sum_i a_i x_i^2 = C \in \mathbb{R},$$

then every x_i is an ordinary integer. In particular, an omnific point on a sphere of ordinary finite radius is an ordinary integer point.

Proof. Each $a_i x_i^2$ is nonnegative and at most C . If $C < 0$ there is no solution; otherwise this gives an ordinary real bound on each $|x_i|$. Use [proposition 2.5](#). $\square$

Corollary 10.3. *The orthogonal matrices over A are exactly the ordinary signed permutation matrices. Thus $O_n(A)$ is the finite group of order $2^n n!$, and the determinant-one subgroup consists of its signed permutation matrices of determinant one.*

Proof. For an orthogonal matrix, the sum of the squares in each row is 1. Every entry is therefore ordinary. An integer row of squared norm one has exactly one nonzero entry, equal to 1 or -1 . Orthogonality of different rows forces those entries into different columns. Conversely, each signed permutation matrix is orthogonal. $\square$

By contrast, $\mathrm{GL}_2(A)$ contains all the infinite unipotent matrices

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad t \in A.$$

The Euclidean orthogonal group is rigid because of positivity, not because all matrix groups over A are finite or constant.

Corollary 10.4 (Symmetric matrix rigidity). *Let M be a symmetric matrix over A . If $\mathrm{tr}(M^2)$ is an ordinary real constant, then every entry of M is an ordinary integer. In particular, this holds if its characteristic polynomial has ordinary integer coefficients.*

Proof. For symmetric M ,

$$\mathrm{tr}(M^2) = \sum_i m_{ii}^2 + 2 \sum_{i < j} m_{ij}^2,$$

so [proposition 10.2](#) applies. The first two characteristic coefficients give $\mathrm{tr}(M^2) = (\mathrm{tr} M)^2 - 2e_2(M)$ when the size is at least two; for size one the assertion is immediate. Hence a constant characteristic polynomial makes this trace constant. $\square$

10.4 Representative equations

Equation or constraint	Exact conclusion over A
$2x + 3y = 1$	$x = -1 + 3t, y = 1 - 2t, t \in A$.
$x^2 + y^2 = N \in \mathbb{Z}$	Only ordinary integer solutions.
$x^2 - Dy^2 = c, Dc \neq 0$	Only ordinary integer solutions.
$x^2 - Dy^2 = 0, D > 0$ nonsquare	$(x, y) = (\pm\sqrt{D}t, t), t \in \mathcal{J}$.
$x^4 - y^6 = c \neq 0$	Only ordinary integer solutions.
$x^2 - y^2 - z^2 = 1$	Infinite unimodular family from (7.2) .
$x^2 + y^2 = z^2$	Infinite unimodular family from proposition 9.4 .
$x^n + y^n = z^n, n \geq 2$	Positive infinite solutions in (9.1) ; these are not primitive.
$y^2 = \omega^2 + 1$	No solution, despite consistent constant and leading tests.
$Q^\top Q = I$	Q is an ordinary signed permutation matrix.

11 Fixed Hahn workspaces and possible formalization

11.1 Which results are local in the exponent group?

For a set-sized ordered abelian group Γ , work in a Hahn field whose monomials are written ω^γ with reverse-well-ordered supports, and define

$$A_\Gamma = \mathbb{Z} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}, \quad B_\Gamma = \mathbb{R} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}.$$

The displayed sums mean elements of the ambient Hahn field with the specified support restrictions. This notation is a convention, not an identification of every abstract group element with a surreal exponent. When an appropriate ordered group embedding into $\mathbf{No}$ has been fixed, it gives the corresponding surreal normal-form workspace.

The retraction, finite quotients, equational transfer, Smith criterion, constant-product rigidity, norm rigidity, quadratic constructions, and quartic guard all apply to A_Γ . Their proofs use finite algebra, leading exponents, and real coefficients, not the availability of a new exponent outside Γ . If $\Gamma \neq 0$, its positive cone provides nonconstant parameters for the infinite-family statements. The case $\Gamma = 0$ is simply $A_\Gamma = \mathbb{Z}$.

By contrast, the common-divisor theorem for an arbitrary set of purely infinite elements and the monomial clearing theorem for an arbitrary set of full-field elements require bounds that may not exist in Γ . For instance, in $\Gamma = \mathbb{R}$, the set $\{1/k : k \geq 1\}$ has no strictly positive real lower bound. In the full surreal group it does.

One must therefore distinguish a proof that stays inside a fixed workspace from a proof that enlarges it. In particular, the equality $\text{Frac}(A) = \mathbf{No}$ and the no-set-of-generators statement for $\mathcal{J}$ are full-class conclusions. They are not asserted for every A_Γ .

11.2 A dependency-oriented formalization plan

The repository already has normal-form and Hahn algebra infrastructure, coefficient extraction, and a carefully stated boundary between concrete surreal results and workspace results [10]. The following is a *proposed* extension plan. The names below are suggested module names, not claims that corresponding files or declarations already exist.

Proposed module	Main obligation and dependency
<code>Omnific.Basic</code>	Define the nonnegative-exponent subring and the integer-constant subring; prove the retraction and finite-element criterion.
<code>Omnific.Residue</code>	Prove $nA = \mathcal{J} + n\mathbb{Z}$, finite quotients, and preservation of $\mathcal{J}$.
<code>Omnific.Rigidity</code>	Prove units of the real and complex support rings are constants; derive product, binary-form, and norm rigidity.
<code>Omnific.Quadratic</code>	Verify the finite bilinear identity for T_t , the isotropic-vector choices, and the level classification.
<code>Omnific.IntegrityGuard</code>	Combine Pell rigidity, ordinary Pell residues, and an explicitly imported three-square theorem.
<code>Omnific.Global</code>	Use the genuine small-cut theorem for support bounds, common monomial divisibility, and global denominator clearing.

A useful separation is to prove the finite algebraic identities first over arbitrary suitable fields, then instantiate them in a normal-form model. The transvection identity, for example, is finite bilinear algebra; it has no dependence on surreal birthday recursion. The only difficult number-theoretic input to the guard is the classical three-square theorem, not an unproved claim about omnific factorization.

No Lean files are included here because this manuscript has not been formalized or checked

against a built repository. A proposed module map must not be confused with a completed formal verification.

12 Further questions and conclusions

12.1 Questions not settled by this article

The following are precise continuation problems. They are not all claimed to be open in the literature; the point is to isolate what the present proofs do not decide.

Primitive higher-degree Fermat fibers. The scaled solutions in (9.1) are never primitive. For a fixed $n \geq 3$, classify primitive or unimodular omnific solutions of $X^n + Y^n = Z^n$ with all coordinates nonzero. The binary nonzero-fiber theorem does not directly answer a three-variable homogeneous question.

Coprime exponent equations and affine curves. Determine which curves such as $Y^2 = X^3 + 1$ have genuinely infinite omnific points. The leading equation gives only a necessary condition; it does not decide whether an algebraic expansion has an inadmissible negative tail. A satisfactory criterion should separate polynomial parametrization, torus rigidity, and genuinely algebraic branches.

The arithmetic of the pure-infinite ideal. We have given a first-order definition of $\mathcal{J}$, but not an existential one. Is there an existential definition over $\mathbb{Z}$? More generally, which fibers or subideals of $\mathcal{J}$ are Diophantine? Retraction stability is necessary for such definitions but far from sufficient.

Complexity of the integrality guard. Can one reduce the number of auxiliary variables below five, retain a single quartic equation, or prove a meaningful lower bound? The present construction establishes an explicit upper bound only. The role of the three-square theorem suggests testing other rigid arithmetic families as finite bounds.

Infinite coefficients and effective fragments. For polynomials with finitely represented omnific coefficients, what can be decided about roots in a chosen support class? The equation $Y^2 = \omega^2 + 1$ shows why a solver needs more than constant and leading conditions. Any effective claim must specify coefficients, exponent representations, and whether workspace enlargement is permitted.

12.2 Synthesis

The arithmetic of A is governed by three simultaneous facts:

$$A/\mathcal{J} = \mathbb{Z}, \quad A^\times = \{\pm 1\}, \quad \text{Frac}(A) = \mathbf{No}.$$

The first makes ordinary congruences and equation-existence questions look classical. The second imposes powerful rigidity on nonzero product fibers. The third makes projective representation extraordinarily flexible, while saying little about primitive representatives.

The difference between a Pell conic and a three-variable indefinite quadric is therefore structural, not a matter of choosing a sufficiently large number. Binary factorization forces

constants; an additional indefinite dimension supports finite polynomial isometries. The quartic guard uses the rigid side of this dichotomy to recover ordinary integers from the much larger omnific ring. These mechanisms provide a workable starting theory of omnific–Diophantine geometry without assuming unique factorization, a Euclidean termination principle, or full transfer from ordinary arithmetic.

A Proof dependencies and verification boundaries

The foundation imported from surreal theory is the existence, uniqueness, order, and multiplication of set-supported Conway normal forms, the set-cut property, and real closedness of **No**. All other main deductions are proved from these and ordinary finite algebra, except where an external number-theoretic theorem is explicitly cited.

The principal dependency chains are:

normal forms $\implies$ `ct` is a retraction $\implies$ equational transfer and finite quotients;
 normal forms $\implies$ constant units in $\mathcal{B}_{\mathbb{C}}$ $\implies$ binary and norm rigidity;
 Pell rigidity + ordinary three-square theorem $\implies$ quartic integrality guard;
 real quadratic linear algebra + finite support-ring algebra $\implies$ quadratic-level dichotomy;
 normal forms + set-cut bounds $\implies$ global common divisors and denominator clearing.

The included `verification.py` checks the cubic norm identity, Pythagorean and hyperboloid identities, Bézout certificates, finite matrix instances of the quadratic isometry identity, Pell residue patterns, and finitely many three-square guard witnesses. It is a supplement, not a proof of statements involving all normal forms or all ordinary integers. It does not implement the full class of surreal numbers.

The paper’s claims have been reviewed by algebraic derivation and the finite checks just listed. No independent referee report or proof-assistant verification is claimed.

B Repository and source audit

B.1 Repository snapshot

The inspected repository was `VladimirReshetnikov/Surreal`, pinned to

`2cb9c0200af749fbbd27796c97d017e7f16fbf33`

The GitHub connector was used to retrieve the repository tree, root README, documentation index, trigonometry directory listing, and `21-period-arithmetic-source_audit.md`. The inspected documentation already discusses normal forms, integer parts, coefficient extraction, Hahn workspaces, and finite residue arithmetic. These are credited as background rather than presented as discoveries of this article.

This was a focused source inspection, not a repository checkout, build, or exhaustive review of every manuscript and Lean declaration. A missing search match in a large returned tree is not treated as proof that the repository contains no related result. The article’s mathematical proofs stand on their stated hypotheses, not on an inference from an incomplete repository search.

B.2 Primary mathematical sources

The 2024 L’Innocente–Mantova article was inspected through its arXiv text, with its opening PDF page also checked as an image. Its normal-form conventions and distinction between ordinary factorization and generalized series factorization were especially relevant. Ehrlich–Kaplan’s initial integer-part lemma and Jeřábek’s residue-kernel discussion were checked in the available article text.

For the classical three-square theorem, the primary mathematical paper [8] and the Archive of Formal Proofs entry [9] were consulted. For undecidability, the author-hosted text accompanying Matiyasevich’s monograph was consulted [7]. The standard books of Conway and Gonshor are background references; they were not newly read in full for this task.

B.3 A dated factorization announcement

A September 18, 2026 announcement reports a Lean proof of Conway’s refinement conjecture [11]. Its author explicitly qualifies the independent-verification status. We did not audit that code or use the announced result. Accordingly, this article does not repeat older literature’s description of the refinement conjecture as a verified *currently open* problem, and does not claim that this manuscript settles it. None of our proofs depends on its present status.

B.4 Novelty and attribution

The strongest proposed contributions of this manuscript are the explicit quadratic-level classification, the uniform five-auxiliary-variable quartic guard, and the organization of rigidity and primitive-point obstructions into an omnific–Diophantine framework. Familiar mechanisms such as coefficient retractions, unit arguments, polynomial parametrizations, and nilpotent quadratic isometries are not claimed as new inventions. A focused search and source review do not establish historical priority; that would require broader expert review.

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